

ZEFF

Z-Score Estimator for Fantasy

How it works: data, formulas, and what we learned testing it

*A plain-English guide to every scoring formula in the tool,
including what we tried and rejected along the way.*

1. What Is ZEFF?

ZEFF is a tool for ranking NBA players ahead of a fantasy basketball draft. Instead of just looking at points per game or a single headline stat, it uses a method called "z-scores" to measure how good a player is across every category that matters in a standard 9-category fantasy league, then layers in real-world context on top: injuries, trades, rookie translations from college stats, and how the real draft market is actually valuing each player.

This document explains, in plain English, every formula the tool uses and why it works the way it does -- including the ideas we tried that did NOT work, and how we found that out. Nothing here is theoretical: every formula described below was tested against real, historical NBA outcomes before it was kept.

2. Where the Data Comes From

ZEFF pulls from five different public sources:

- stats.nba.com -- season stats (points, rebounds, shooting, etc.) and team pace.
- Basketball-Reference -- the season's trade and transaction history.
- ESPN -- the current, live injury report.
- sports-reference.com/cbb -- college stats, used to estimate rookies who don't have NBA stats yet.
- hashtagbasketball.com -- real average draft position (ADP) from actual Yahoo, ESPN, and Fantrax fantasy drafts.

A technical note worth including: several of these sites actively try to block automated requests. We get past that the same way many legitimate data tools do -- by making our requests present themselves like an ordinary web browser, a technique called TLS fingerprint impersonation. One source we looked at for historical injury records blocks even that -- it uses a security challenge that only a real, human-operated browser can solve -- so we don't use it. Section 8 covers what we tried and chose not to pursue.

3. The Core Ranking: 9-Category Z-Scores

3.1 What a z-score means

A z-score answers one question: how far above or below average is this player in a given category, measured in a way that lets us fairly compare categories that live on completely different scales (points versus blocks versus a shooting percentage)? A z-score of 0 means exactly average for the player pool. +1 is solidly above average. +3 is elite -- among the very best in the league. Negative numbers mean below average.

We score the 9 categories used by most standard fantasy leagues: Points, Rebounds, Assists, Steals, Blocks, Three-Pointers Made, Field Goal %, Free Throw %, and Turnovers.

3.2 The formulas

For most categories (points, rebounds, assists, steals, blocks, three-pointers):

$$z = (\text{player's stat} - \text{pool average}) / (\text{pool spread})$$

Two categories need special handling.

Shooting percentages (FG%, FT%)

A player who is 3-for-3 has a "perfect" percentage, but that's not meaningful. So instead of scoring the raw percentage, we score how much value a player's shooting actually adds or subtracts, based on real shot volume:

$$\begin{aligned} \text{impact} &= (\text{player's \%} - \text{pool average \%}) \times \text{player's attempts} \\ z &= (\text{impact} - \text{average impact}) / (\text{spread of impact}) \end{aligned}$$

This way, a high-volume shooter with a good percentage is worth far more than a bench player who barely shoots, even if their raw percentages look similar.

Turnovers

Fewer turnovers is better, so the sign gets flipped -- a player who turns the ball over a lot gets a NEGATIVE score here, not a positive one.

All 9 category z-scores are added together into one "Total Z" number per player, and that total is the ranking.

3.3 Why only the top 200 players?

In a typical 12-team fantasy league, only around 200 players actually get drafted. If we calculated the "average" and "spread" for each category using the entire NBA (450+ players), a large group of bench

players who will never be drafted would drag those numbers around and distort what actually matters on draft day.

So the ranking works in two passes: first, it ranks the whole eligible player pool to identify who the top 200 are. Then it throws out everyone else and re-runs the entire z-score calculation using ONLY those 200 players as the comparison group. This makes every number reflect real draft-day value instead of generic, whole-league value.

4. Injury Risk Score

We wanted one number that answers "how much should I worry about this player's health and reliability?" We looked for a proper historical injury database (the kind real analysts use) but the standard source for that blocks automated access even with a real browser (see Section 8). So instead, we built this score from four things we DO have solid data for, multiplied together:

Factor	What it measures	Formula
Age	Older players carry more risk.	No penalty under 24. Above 24: risk grows 5% for every year of age.
Durability history	Games missed relative to the player's own team, averaged over recent seasons.	Risk scales up with the average fraction of team games missed.
Current status	Is this player listed as injured right now?	"Out" adds the most risk (1.5x), "Day-to-Day"/"Questionable" a little (1.1x), healthy players get no penalty (1.0x).
On-court role	A bigger role means a bigger loss if the player gets hurt.	Scales the whole score by minutes played per game, out of a 36-minute full-time benchmark.

These four factors are MULTIPLIED together, not averaged. That means if any single one of them is a real concern, the whole score moves meaningfully -- it doesn't get diluted just because the other three factors look fine.

Important honesty check: this is a proxy built from missed games, age, and current status -- it is not a record of actual injury diagnoses. We say this explicitly on the tool itself so it's never mistaken for medical information.

5. Translating College Stats for Rookies

Brand-new rookies don't have NBA stats yet, so a normal ranking would leave them off the board entirely -- even ones who are obviously going to matter. For any named rookie, ZEFF can pull their final college season and convert it into a rough "if they played in the NBA" estimate, using simple discount factors:

Stat	Adjustment	Why
Points, rebounds, assists, steals, blocks, threes, shot volume	Scaled to 75%	College competition and pace are different from the NBA.
Shooting percentages	Scaled to 93%	Shooting touch tends to carry over better than raw production does.
Turnovers	Scaled UP to 115%	Rookies cough it up more against NBA-level speed and length.
Minutes played	Scaled to 85%	Rookies rarely walk into a full college-sized role immediately.

This is explicitly a rough, hand-built estimate -- not a real statistical projection. Every player added this way is clearly labeled on the board (a "Rookie" tag) so it's never confused with real, played NBA production.

6. The Forward-Looking Projection

Everything above describes what a player has ALREADY done. This section covers the part of the tool meant to answer a harder question: what will this player do NEXT season? We spent the most time here, and it's worth walking through honestly, including the parts that didn't work, because that process is exactly how we ended up trusting the version we shipped.

6.1 How we tested whether a formula actually works

We have several years of real, already-recorded NBA stats. That let us test any projection idea properly: pretend we're standing at the start of a season we already know the outcome of, run the formula, and check whether it actually predicted what really happened. We did this across three separate season-to-season jumps, so we weren't just getting lucky on one.

Every idea was compared against the simplest possible prediction: assume nothing changes, and next season looks exactly like last season. This is the "naive" baseline. If a more complicated formula can't beat that simple guess, it isn't actually adding value, no matter how sophisticated it sounds.

6.2 What we tried first, and why it failed

The first version of the projection blended the last three seasons together (weighting recent seasons more), nudged unusually hot or cold shooting seasons back toward normal, and applied an age adjustment that boosted young players and penalized older ones.

When tested, it did not beat the naive baseline -- and in some tests it did worse. Specifically:

- Blending multiple seasons together with a fixed weighting hurt accuracy every time. It was diluting good, recent, reliable data with older data that mattered less.
- The shooting-percentage correction didn't clearly help either way.
- The age penalty for older players actually made things worse, which surprised us and led directly to the fix described below.

6.3 What we validated and now use

We went back and tested each idea separately, refined the promising ones, and always checked the result on data the formula had never seen while being tuned -- the same discipline as a proper experiment: tune on one set of seasons, test on a different one you didn't touch. Two ideas survived this process.

A smarter way to combine seasons

Instead of blending three seasons with a fixed formula, we blend just the last two seasons, weighting each one by how many games that player actually played that season:

```
blended stat = (stat_lastyear x games_lastyear + stat_prioryear x games_prioryear)
               -----
               (games_lastyear + games_prioryear)
```

A healthy player who played a full season is barely affected by this -- their most recent season is trusted almost completely. But a player who was hurt and only played a handful of games automatically pulls in more weight from their healthier prior season, which is a fairer way to handle an injury-shortened year. This beat the simpler "just use last season" approach in all three of our tests, including the one it was never tuned on.

A corrected age curve

Young players (under 25) still get a production boost that grows the younger they are -- 2.5% more production for every year under 25 -- which makes sense, since young players are still developing into their NBA role. But we removed the penalty for older players entirely, because testing showed it was actively making predictions worse.

```
age factor = 1.0 + (max(0, 25 - age) x 2.5%)
(no additional adjustment, positive or negative, for players 25 or older)
```

Our best guess as to why: a player's most recent season already reflects any real decline that happened -- an older player who has slowed down already shows a lower stat line because of it, so an extra age penalty on top was double-counting the same effect. It's also likely that the players who remain relevant enough to stay in a 200-player draft pool are, by definition, the ones who tend to be aging gracefully -- the ones who didn't tend to have already fallen out of the league entirely.

6.4 What we tried after that, and rejected

We also tested a "momentum" idea: if a player's stats were trending up or down between two seasons, assume some of that trend continues. This did not hold up. The "best" setting for it flipped from positive to negative depending on which seasons we tested it against -- a textbook sign that a formula is fitting noise in the data rather than a real, repeatable pattern. It was left out of the final formula.

6.5 Honest accuracy summary

Here is what "beating naive" actually looked like, measured by rank correlation (did we get the order right?) across the three season-transitions we tested:

Transition	Naive	Age curve only	Age curve + blend
2022-23 -> 2023-24	0.754	0.769	0.784 (best)
2023-24 -> 2024-25	0.702	0.736	0.758 (best)
2024-25 -> 2025-26	0.681	0.712	0.732 (best)

Every column beats the one before it, in all three transitions, with no exceptions. The age curve constants were grid-searched on the first two transitions only; 2024-25 -> 2025-26 was held out of that search and only scored afterward -- the final formula (age curve + games-weighted blend) still won there too. That is a real, if modest, improvement -- not a guarantee. It should be treated as a useful signal on the board, not a certainty. A large gap between a player's historical and projected ranking is worth noticing, not blindly trusting.

6.6 A note on trades

When a player gets traded, we deliberately do NOT try to guess how their role will change on the new team -- nobody can reliably predict that from stats alone; it depends on coaching decisions and roster fit that box scores can't see. What we DO adjust for is something we can actually measure: the new team's pace. A faster-playing team creates more possessions for everyone on the floor, which mechanically means more scoring, rebounding, and assist opportunities regardless of role. So a traded player's counting stats are scaled by the new team's pace relative to the old team's, and nothing more.

7. Comparing to the Real Draft Market (ADP)

"ADP" stands for Average Draft Position: literally, on average, what pick number a player gets taken at across real fantasy drafts. We pull this from hashtagbasketball.com, which blends ADP data from Yahoo, ESPN, and Fantrax leagues, and show it right next to our own ranking.

```
vs ADP = ADP - our rank
```

A big POSITIVE number means the real market is drafting a player much later than our numbers say they should go -- a potential value pick, or possibly a sign the market knows something our stats-only model can't see (an off-court concern, a crowded position battle, and so on). A big NEGATIVE number is the reverse: the market is drafting them earlier than our numbers would.

This is deliberately a different KIND of signal than everything else in the tool. It isn't derived from box scores at all -- it's real behavior from real people making real draft picks, which makes it a genuinely independent sanity check against our own numbers.

8. What We Tried That Didn't Work

Being upfront about dead ends is part of being honest about what this tool can and can't do:

- A dedicated historical injury database (the standard source real fantasy analysts use for injury history) blocks automated access behind a security challenge that only a real, human-operated browser can solve. We chose not to try to defeat that.
- We tried pulling professional-analyst consensus rankings through a paid provider's API, but the free tier only returned 10 players -- nowhere near enough to be useful against a 200-player draft board.
- We looked at getting real draft-market data from another well-known provider, but that site loads its data in a way we can't read directly without running the page's own code in a full web browser -- which we decided wasn't worth the added complexity once hashtagbasketball.com turned out to work directly.
- A "momentum" (trend-continuation) signal for the projection formula, covered in Section 6.4 -- tested and rejected for not holding up.

9. Known Limitations, Stated Plainly

- Name matching across different data sources isn't perfect. Occasionally a player can fail to link up between two sources if they're spelled slightly differently.
- Trade detection only automatically figures out simple, two-team trades. Complex, multi-team trades need to be entered by hand.
- The projection is a modest, tested improvement over doing nothing -- not a crystal ball. Treat it as a

tiebreaker between similar players, not as gospel.

- The injury risk score is a proxy built from missed games and current status, not a record of real injury diagnoses.
- A handful of well-known players have occasionally been missing from the ADP data for reasons we haven't fully diagnosed -- worth double-checking if a star's ADP column looks blank.

Appendix: Full Formula Reference

9-cat z-score

Counting stats:	$z = (\text{player} - \text{pool average}) / (\text{pool spread})$
Percentages:	$\text{impact} = (\text{player \%} - \text{pool avg \%}) \times \text{attempts}$ $z = (\text{impact} - \text{avg impact}) / (\text{spread of impact})$
Turnovers:	$z = -1 \times [(\text{player} - \text{pool average}) / (\text{pool spread})]$
Total Z:	sum of all 9 category z-scores
Draft pool:	re-computed using only the top 200 players (2-pass)

Injury risk

```

injury risk = age_factor x durability_factor x severity_factor x impact_factor
age_factor      = 1.0 + max(0, age - 24) x 0.05
durability_fac. = 1.0 + (avg. pct. of team games missed) x 1.5
severity_factor = Out: 1.5 | Doubtful: 1.3 | Day-to-Day/Questionable: 1.1
                  | Probable: 1.02 | none: 1.0
impact_factor   = minutes per game / 36
  
```

College stat translation

```

Counting stats & shot volume:  x 0.75
Shooting percentages:          x 0.93
Turnovers:                     x 1.15
Minutes played:                 x 0.85
  
```

Forward-looking projection

```

Step 1 - blend last 2 seasons, weighted by each season's own games played:
  blended = (stat_t1 x games_t1 + stat_t2 x games_t2) / (games_t1 + games_t2)

Step 2 - age curve (peak age 25, no penalty past peak):
  age_factor = 1.0 + max(0, 25 - age) x 0.025
  
```

Step 3 - pace adjustment for traded players only:
scale by (new team's pace / old team's pace)

Draft-market comparison

vs ADP = ADP - our rank